

Finitely Many Points on the Line Under the Influence of an Exponential Potential — An Integrable System

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ABSTRACT

We analyze the complete integrability of the non-periodic Toda lattice: a one-dimensional system of n point particles interacting via nearest-neighbor repulsive exponential potentials. By introducing a symmetric tridiagonal Lax pair (L, B) , the nonlinear Hamilton-Jacobi equations are mapped to a linear matrix Lax flow $dL/dt = [B, L]$. We establish that as $t \rightarrow +\infty$, the off-diagonal coupling terms decay exponentially to zero, and the diagonal elements (momenta) converge monotonically to the asymptotically decoupled particle velocities ordered strictly by size: $v_1 < v_2 < \dots < v_n$. The asymptotic scattering map thus acts as an intrinsic continuous-time sorting machine.

1. Hamiltonian of the Non-Periodic Toda Lattice

The Toda lattice is defined by the Hamiltonian:

$$H(q, p) = \frac{1}{2} \sum_{k=1}^n p_k^2 + \sum_{k=1}^{n-1} \exp(q_k - q_{k+1})$$

where q_k denotes the position of particle k and p_k denotes its conjugate momentum. The equations of motion are $dq_k/dt = p_k$ and $dp_k/dt = \exp(q_{k-1} - q_k) - \exp(q_k - q_{k+1})$.

2. The Flaschka-Moser Coordinate Transformation

Following Hermann Flaschka (1974), we define the canonical variables:

$$a_k = \frac{1}{2} \exp(q_k - q_{k+1}) \quad (k = 1, \dots, n-1)$$

$$b_k = \frac{1}{2} p_k \quad (k = 1, \dots, n)$$

In these coordinates, the Toda equations take the algebraic polynomial form:

$$da_k/dt = a_k (b_{k+1} - b_k)$$

$$db_k/dt = 2 (a_k^2 - a_{k-1}^2)$$

3. The Tridiagonal Lax Pair

Moser arranged these coordinates into a symmetric Jacobi matrix L and skew-symmetric matrix B :

$$L = \text{tridiag}(a_k, b_k, a_k), \quad B = \text{tridiag}(a_k, 0, -a_k)$$

Then the equations of motion are identically equivalent to the Lax equation:

$$dL/dt = [B, L] = B L - L B$$

Consequently, the eigenvalues $\lambda_1, \dots, \lambda_n$ of L are rigorous first integrals of motion (conserved quantities).

4. Asymptotic Sorting Dynamics ($t \rightarrow \infty$)

As $t \rightarrow \infty$, the particles disperse indefinitely, so $q_{k+1} - q_k \rightarrow \infty$, which drives $a_k(t) \rightarrow 0$ at an exponential rate.

Since all off-diagonal terms a_k vanish asymptotically, $L(t)$ converges to a diagonal matrix:

$$\lim_{t \rightarrow \infty} L(t) = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$$

Because faster particles must eventually pull ahead of slower particles on the real line, the asymptotic velocities are strictly sorted from left to right: $\lambda_1 < \lambda_2 < \dots < \lambda_n$!